FABRIC MANUFACTURING

Part: B

Course Code: FE 253-0723

Weft Knitting Machines

Arnob Mahmud

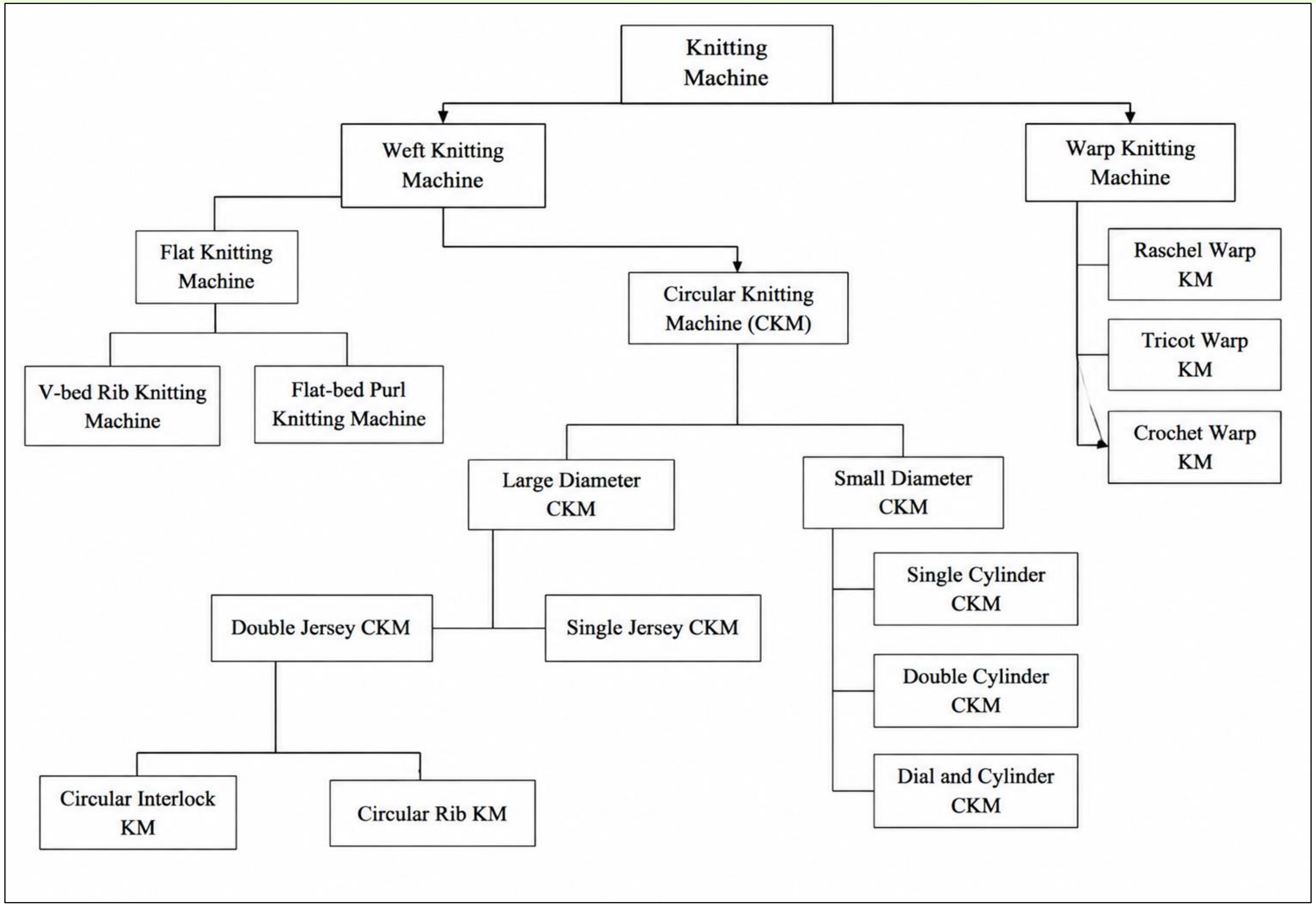
Lecturer

Department of Fabric Engineering

Bangladesh University of Textiles (BUTEX)

E-mail: arnob@fe.butex.edu.bd

CLASSIFICATIONS



TYPES OF MACHINE

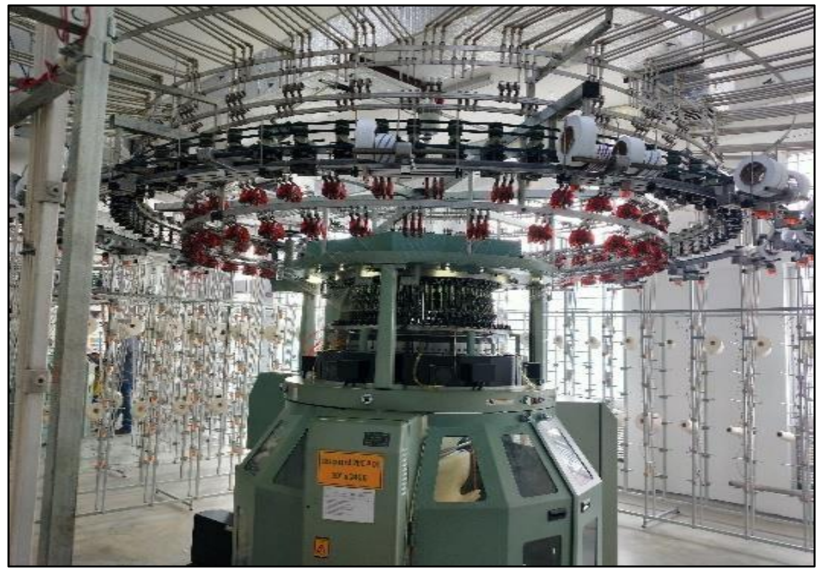
S/J Circular m/c

Fleece Circular m/c

Rib Circular m/c

Interlock Circular m/c

Auto-stripe Circular m/c



S/J Jacquard Circular m/c

D/J Jacquard Circular m/c

Flat Bed Knitting m/c

FEATURES OF CIRCULAR KNITTING MACHINE (CKM)

1. Circular knitting machine has normally rotating (clockwise) cylindrical needle bed.
2. On circular knitting machines mainly latch needles are used.
3. For single jersey machine holding down sinkers are used, one between every needle space. For double jersey machine, dial needle bed replaces the sinker.
4. Normally stationary angular cam systems are used for needle and sinker.
5. Latch needle cylinder and sinker ring or dial revolve through the stationary knitting cam systems.
6. For single jersey machine, sinker ring which is simply and directly attached to the outside top of the needle cylinder.
7. Stationary yarn feeders are situated at regular intervals around the circumference of the rotating cylinder.
8. Yarn is supplied from cones placed either on an integral over head bobbin stand or on a free-standing creel through tensioners, stop motions and yarn guide eyes down to the yarn feeder guides.

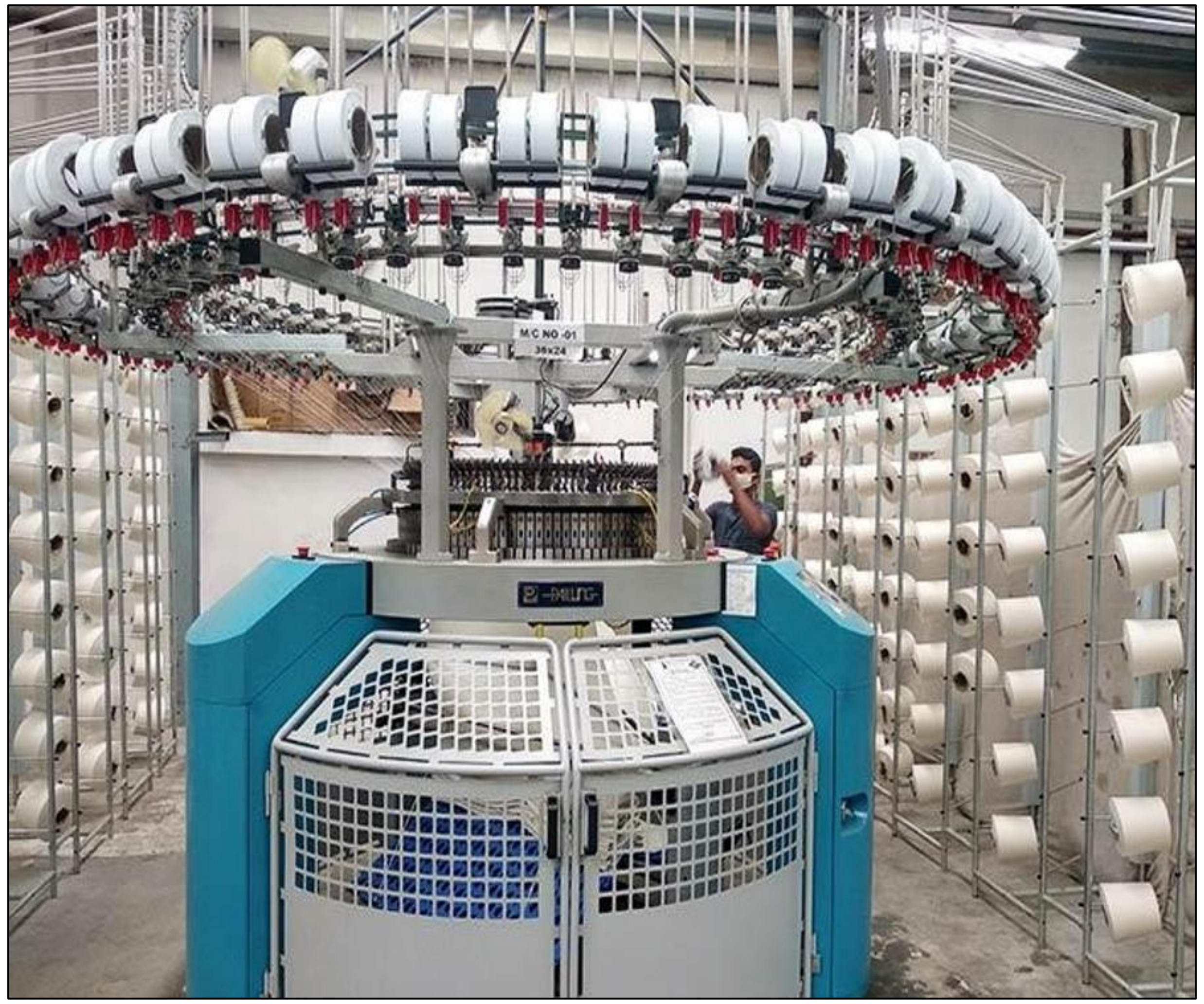
FEATURES OF SJCKM

1. It is a single-bed circular knitting machine having only one set of needles.
2. The needles are latch-type and mounted vertically in the slots of the needle cylinder.
3. The sinkers are fitted between every two needles and move radially.
4. The machine produces plain fabric (single jersey) which has technical face and back sides.
5. Usually, the diameter of the machine ranges from 12 to 60 inches.
6. The gauge may vary from 14 to 36 needles per inch, depending on yarn count and end use.
7. Cam systems are placed around the cylinder to control needle movement for knit, tuck, and miss stitches.
8. The yarn feeders (generally 90-120 per machine) feed yarns continuously to each knitting head.
9. Sinker ring which is simply and directly attached to the outside top of the needle cylinder.
10. The fabric is drawn down through the center of the cylinder by take-down rollers.
11. The production rate depends on machine diameter, speed (revolutions per minute - RPM), and number of feeders.
12. The machine frame holds the main drive, take-down mechanism, fabric collection, and lubrication systems.
13. Fabrics produced are used for T-shirts, underwear, sportswear, and other lightweight garments.

SJCKM PARTS

- Creel
- Cone package holder
- Yarn passing tube
- Tensioner
- Positive feeder
- VDQ pully
- Stop motion device
- Feeder
- Needle

- Cam
- Cam box
- Cylinder
- Sinker
- Fabric spreader
- Take down roller
- Cloth roller
- Lubrication system
- Dust removing fan

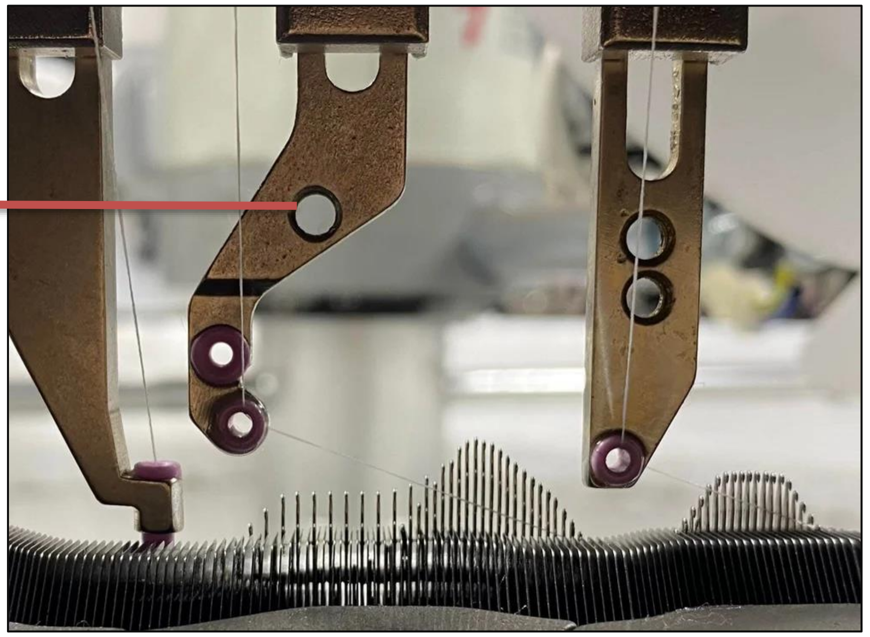


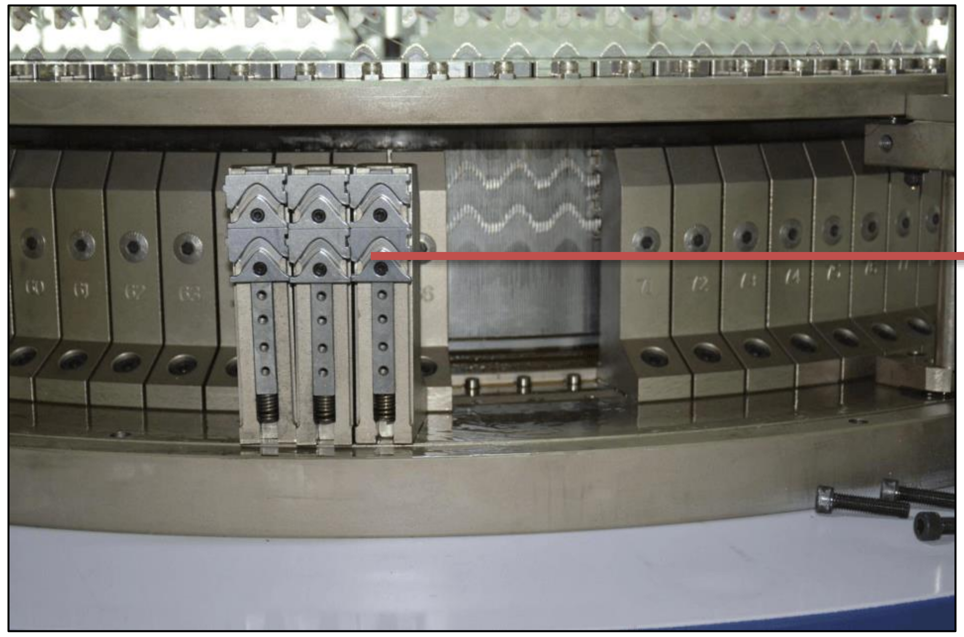
SJCKM PARTS

POSITIVE FEEDER

SJCKM PARTS

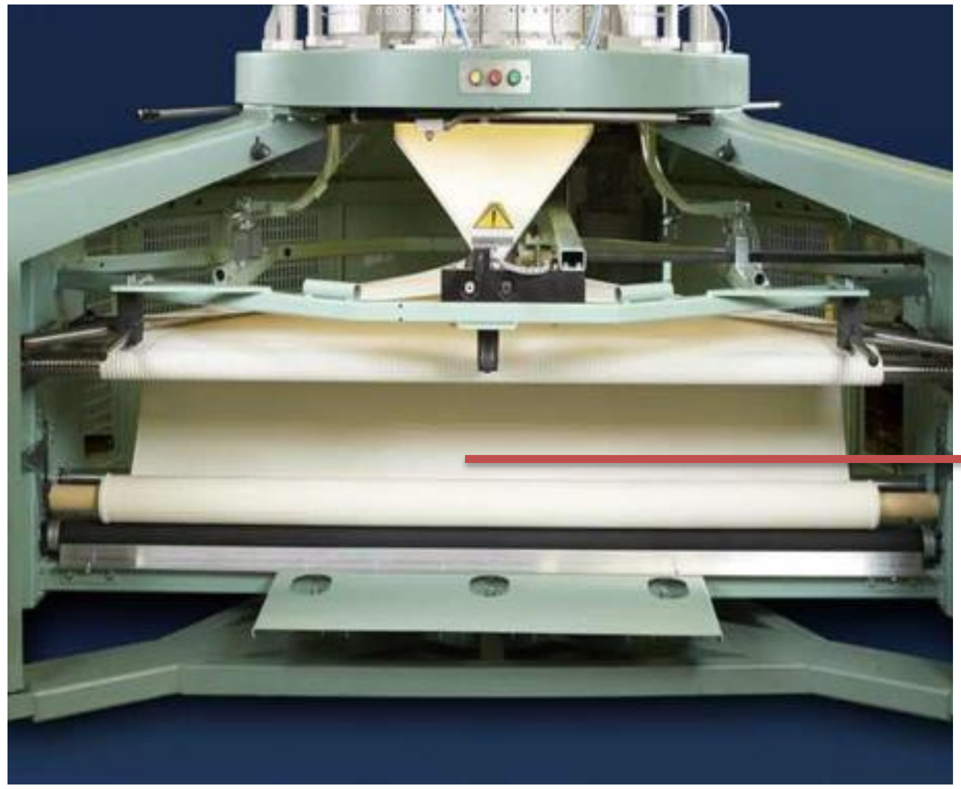
YARN FEEDER





CAM

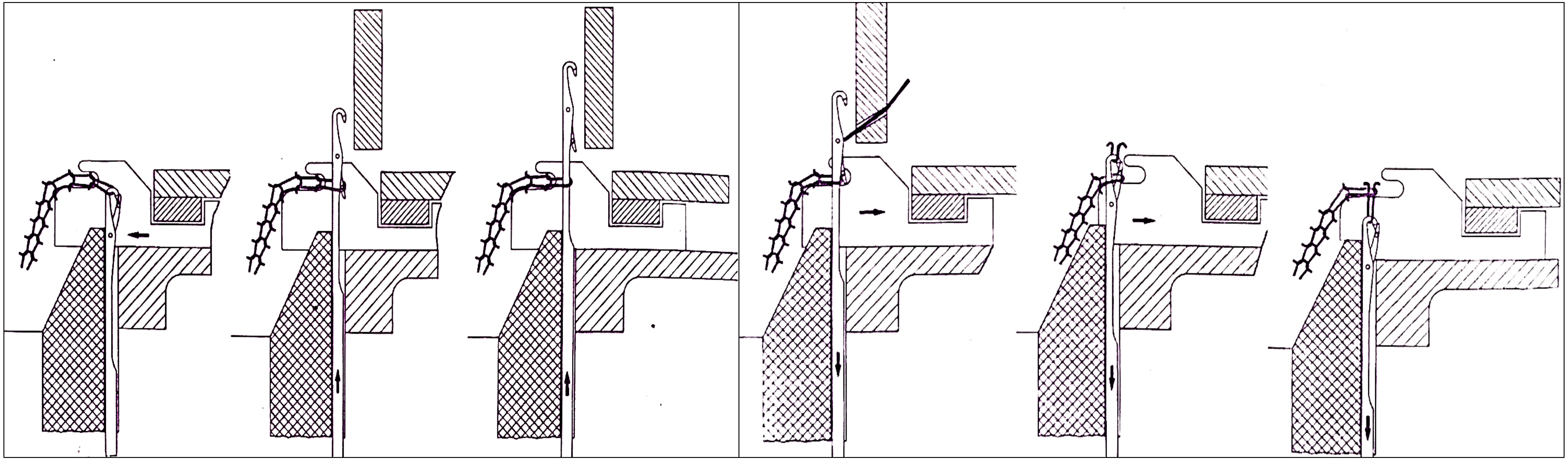
CYLINDER



SPREADER

https://youtu.be/9M_0l3z7x84?si=BguwdVXdVHNCHv7L

KNITTING ACTION OF SJCKM

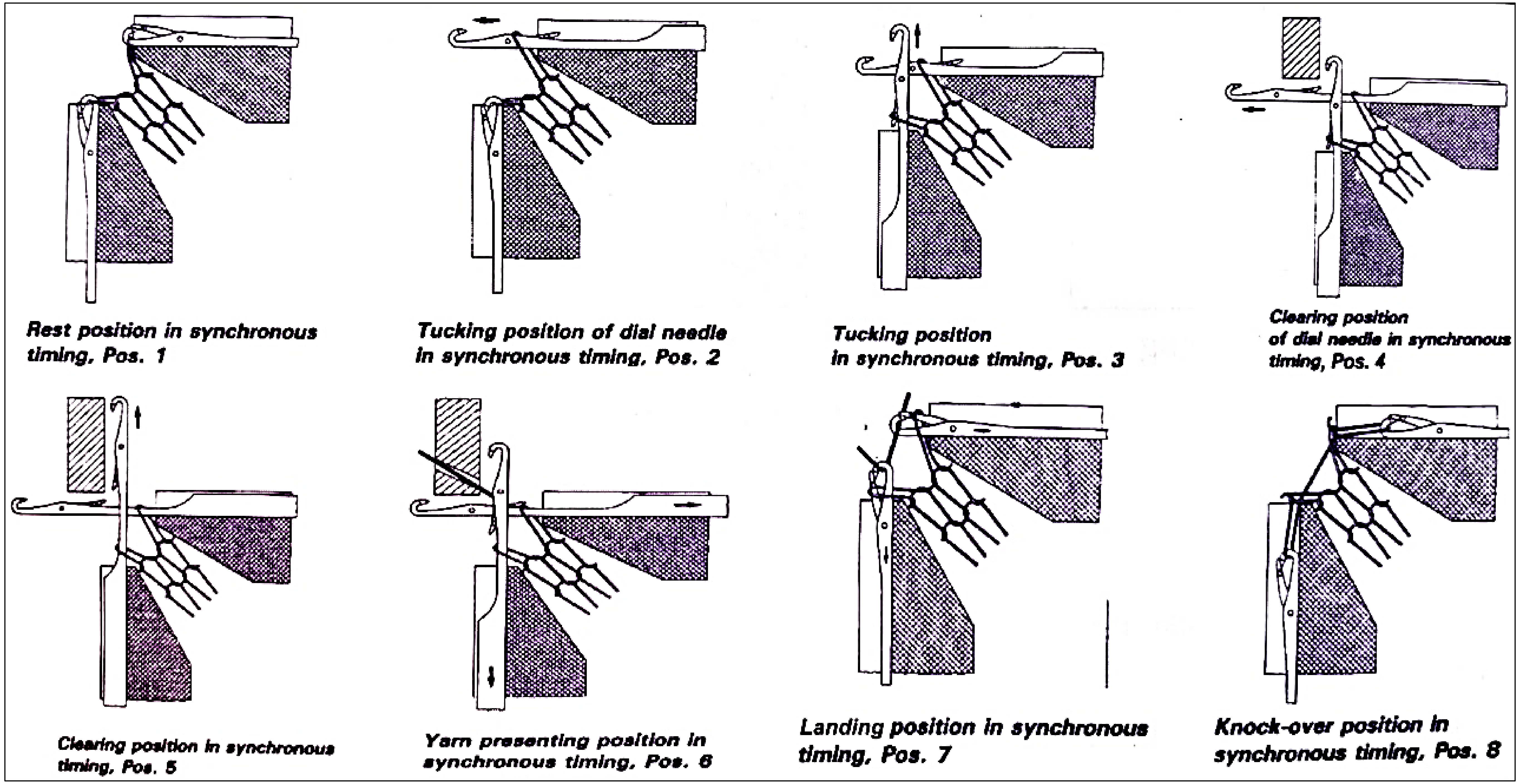


FEATURES OF DJCKM

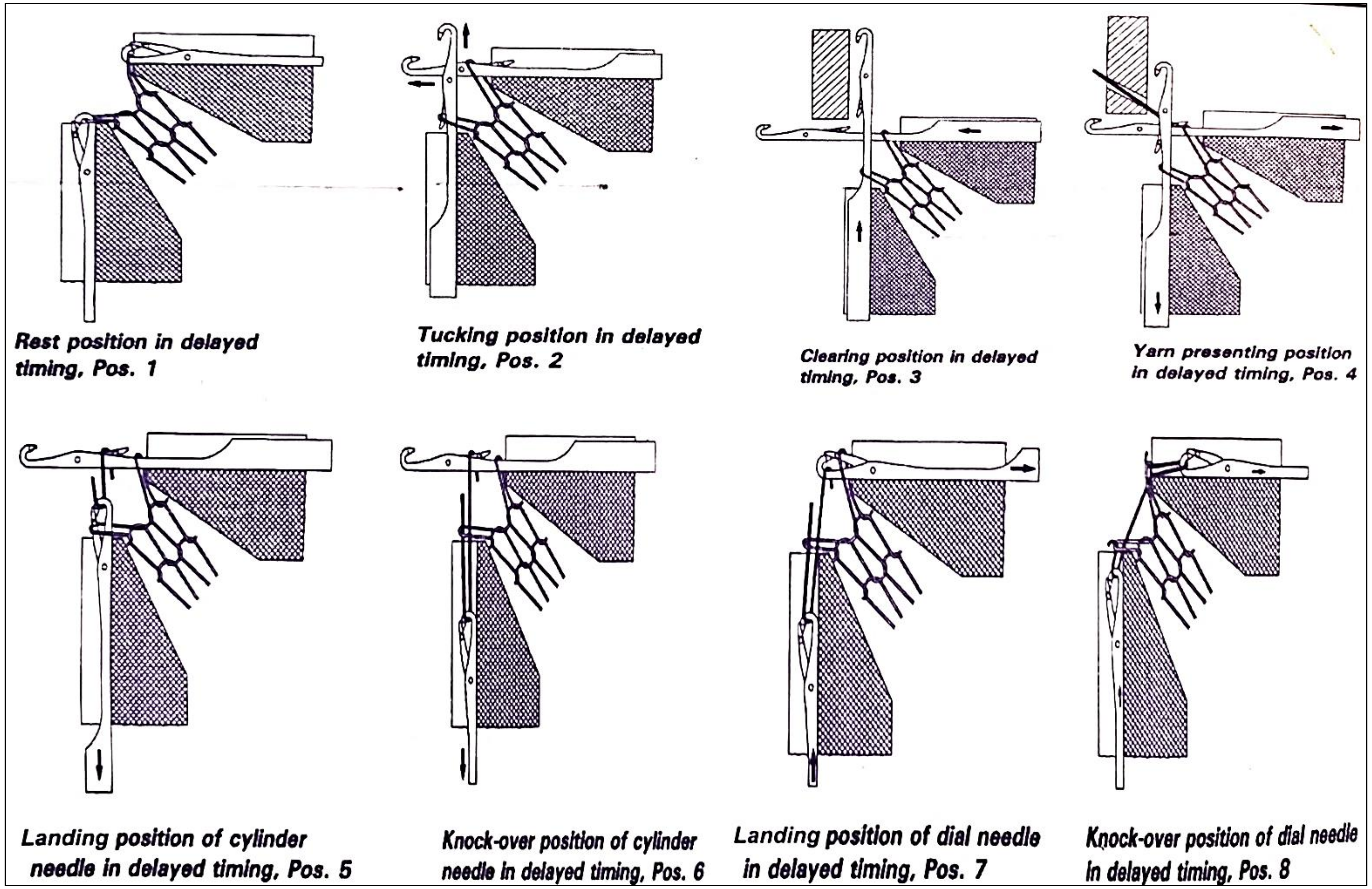
- The circular knitting machine which having two sets of needles; one on cylinder & another one on dial and there are no sinkers here is known as double jersey machine.
- This double arrangement of needles allows the fabric to be manufactured which is twice as thick as the single jersey fabric, known as double jersey fabric.
- There are three types of double jersey circular knitting machines are generally used to produce double jersey fabric. These are:-

  1. Rib circular knitting machine,
  2. Interlock circular knitting machine &
  3. Purl circular knitting machine.

KNITTING ACTION OF DJCKM



KNITTING ACTION OF DJCKM



Rest position in delayed timing, Pos. 1

Tucking position in delayed timing, Pos. 2

Clearing position in delayed timing, Pos. 3

Yarn presenting position in delayed timing, Pos. 4

Landing position of cylinder needle in delayed timing, Pos. 5

Knock-over position of cylinder needle in delayed timing, Pos. 6

Landing position of dial needle in delayed timing, Pos. 7

Knock-over position of dial needle in delayed timing, Pos. 8

FLAT BED KNITTING MACHINE

1. The machine consists of two straight needle beds arranged horizontally and facing each other, forming a V-shape (in a V-bed machine).
2. It is used mainly for producing weft-knitted fabrics such as rib, interlock, and purl structures.
3. Each bed contains latch needles which can be individually controlled for stitch selection.
4. Cams mounted on the carriage move across the needle beds to operate the needles and form loops.
5. The carriage carries the cam system and moves to and fro (bi-directional cam system) across the beds to complete each course.
6. Yarn is fed to the needles through a yarn carrier mounted on the carriage.
7. The machine width is specified in terms of the working width of the needle beds, typically 36 - 100 inches.
8. Gauge (number of needles per inch) may range from 3 to 18 G, depending on fabric type and yarn count.
9. The machine can be hand-driven or power-driven, depending on model and production requirements.
10. Fabric take-down is achieved by rollers or weights pulling the fabric downward.
11. Flat bed machines are well-suited for fashion garments, sweaters, collars, ribs, and fully-fashioned components.
12. The structure of the machine allows easy patterning and individual needle selection by electronic or mechanical means.
13. Stitch cams determine the depth of loop formation, and tuck or miss cams can be used for pattern variation.
14. The machine offers high fabric dimensional stability and good control over fabric width and shape.

FLAT BED KNITTING MACHINE

15. The flat machine is the most versatile of weft knitting machines, its stitch potential includes needle selection on one or both beds, needle-out designs, striping, tubular knitting, changes of knitting width and loop transfer.

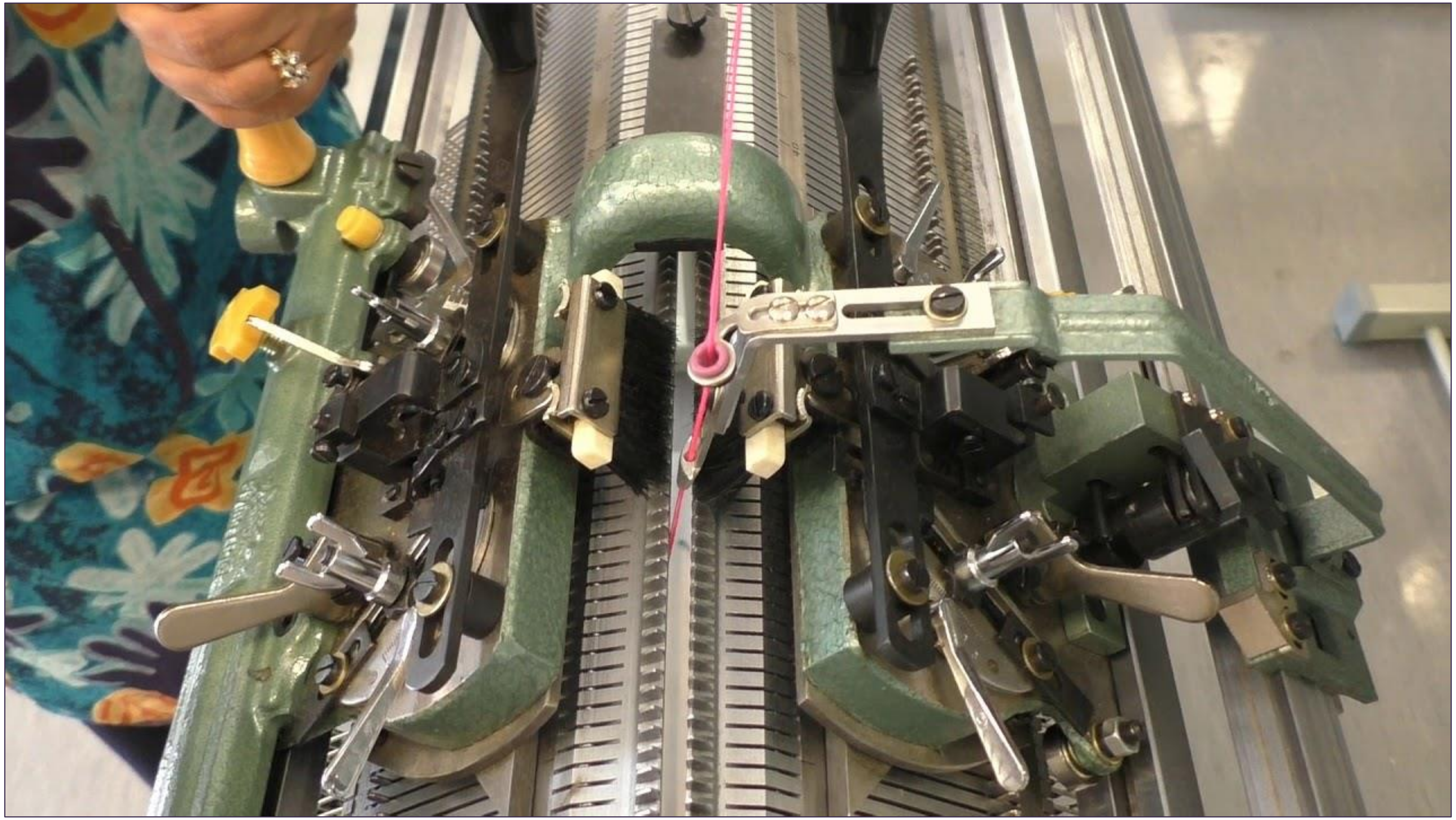
16. A wide range of yarn counts may be knitted per machine gauge including a number of ends of yarn in one knitting system, the stitch length range is wide and there is the possibility of changing the machine gauge.

17. The operation and supervision of the machines of the simpler type is relatively less arduous than for circular weft knitting machines.

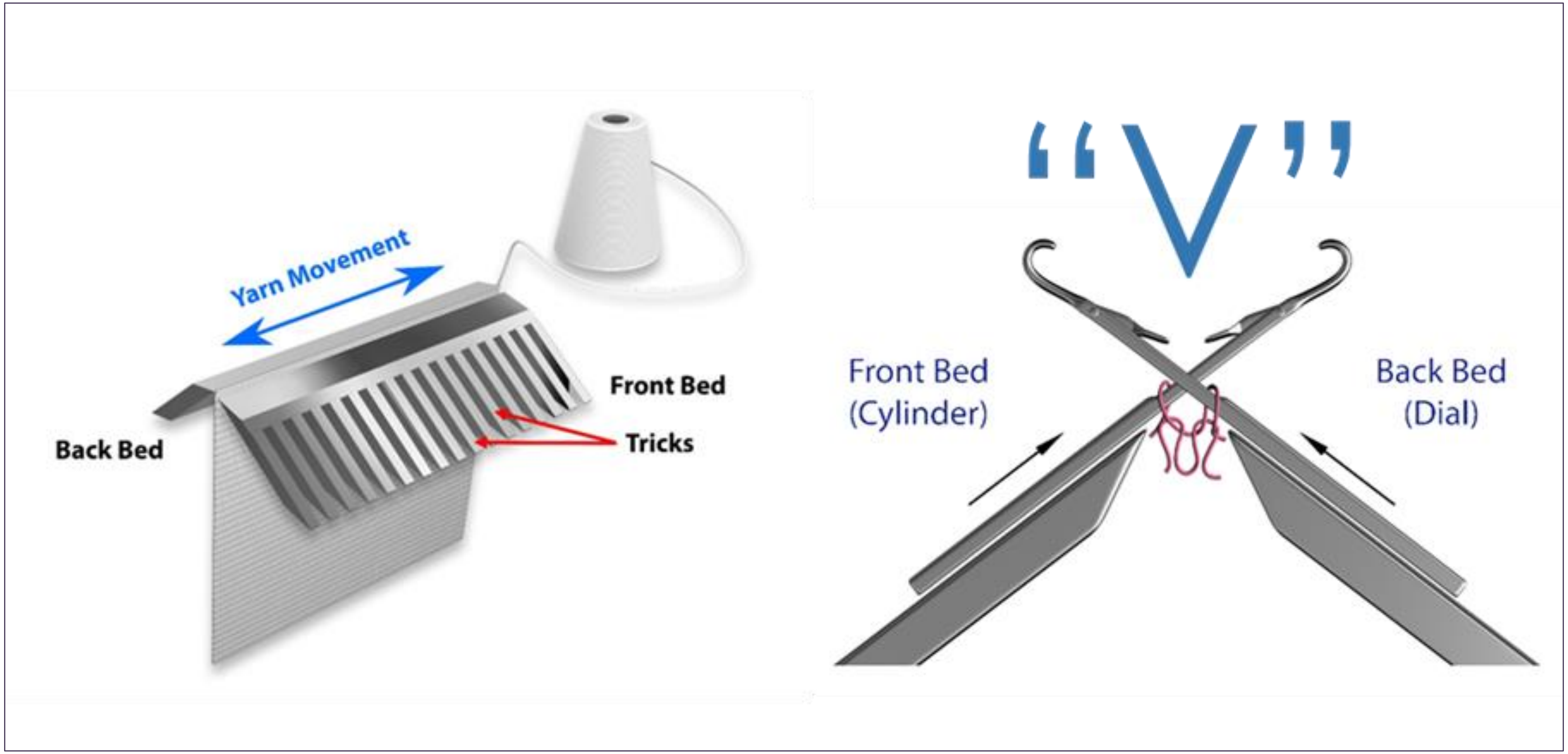
18. The number of garments or panels simultaneously knitted across the machine is dependent upon its knitting width, yarn carrier arrangement, yarn path and package accommodation.

19. Versatile knit structures can be produced by changing the selection of cam carriage.

V BED KNITTING MACHINE



V BED KNITTING MACHINE

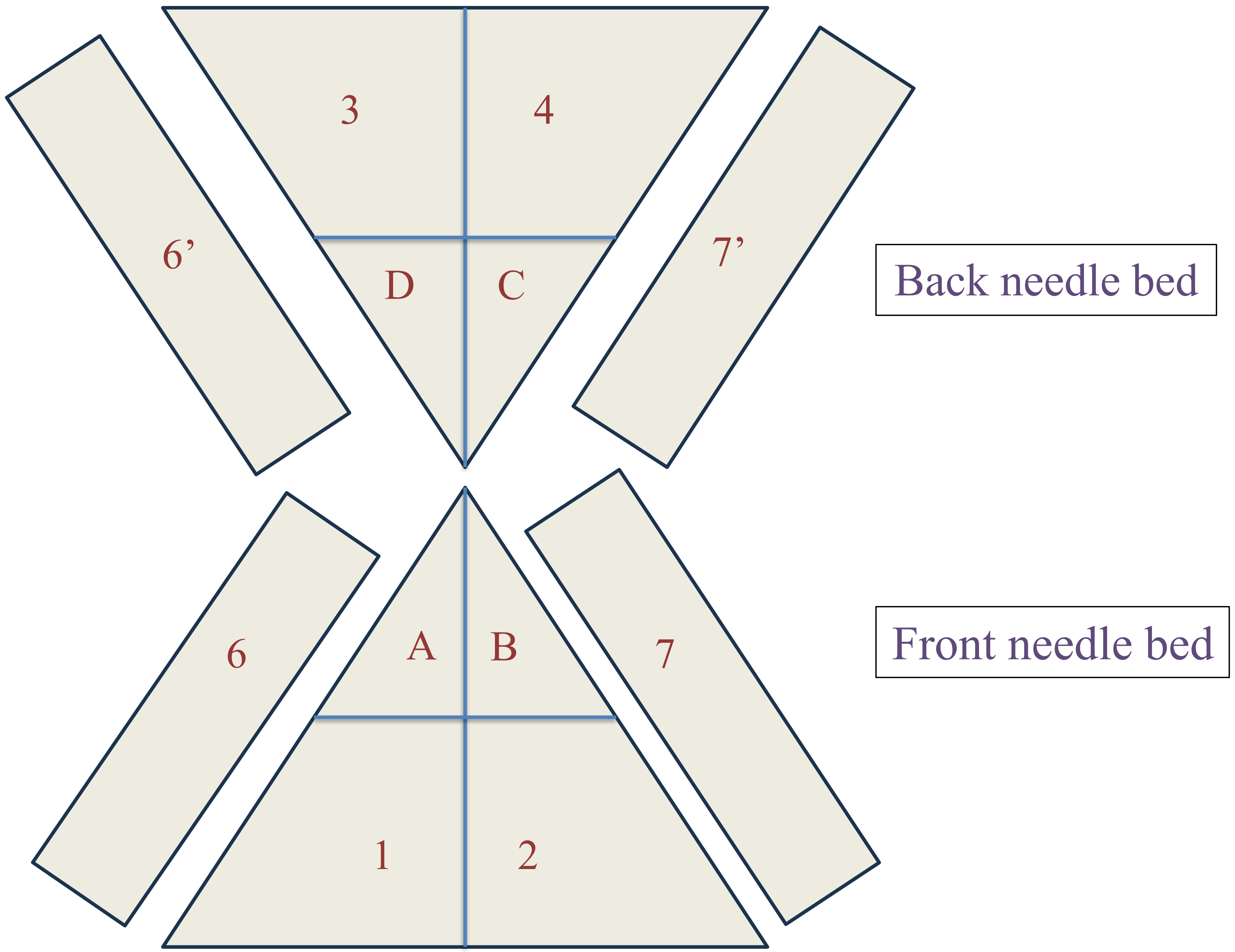


V BED KNITTING MACHINE

1,2,3,4 - Raising Cam

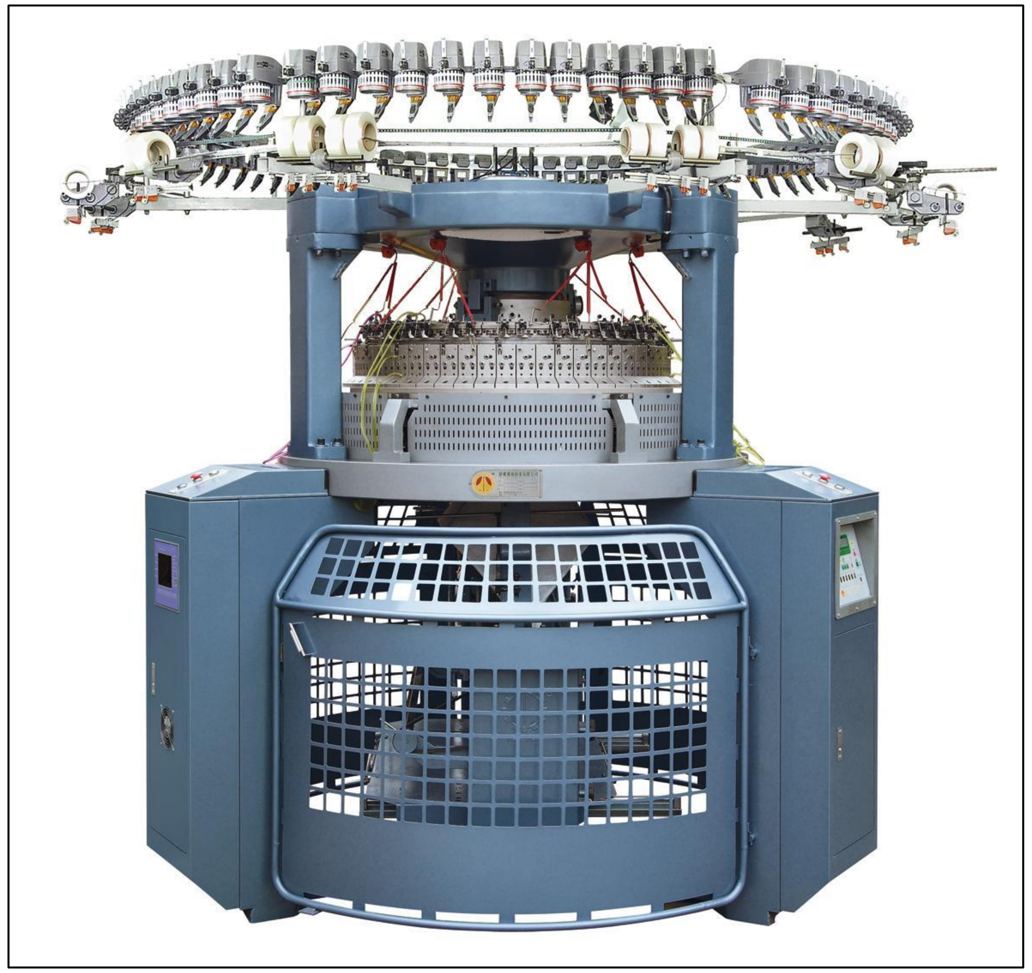
A, B, C, D - Tuck cam/Cardigan cam

6,6’,7,7’- Stich cam



TYPES OF WEFT KNITTING MACHINE BASED ON END PRODUCT

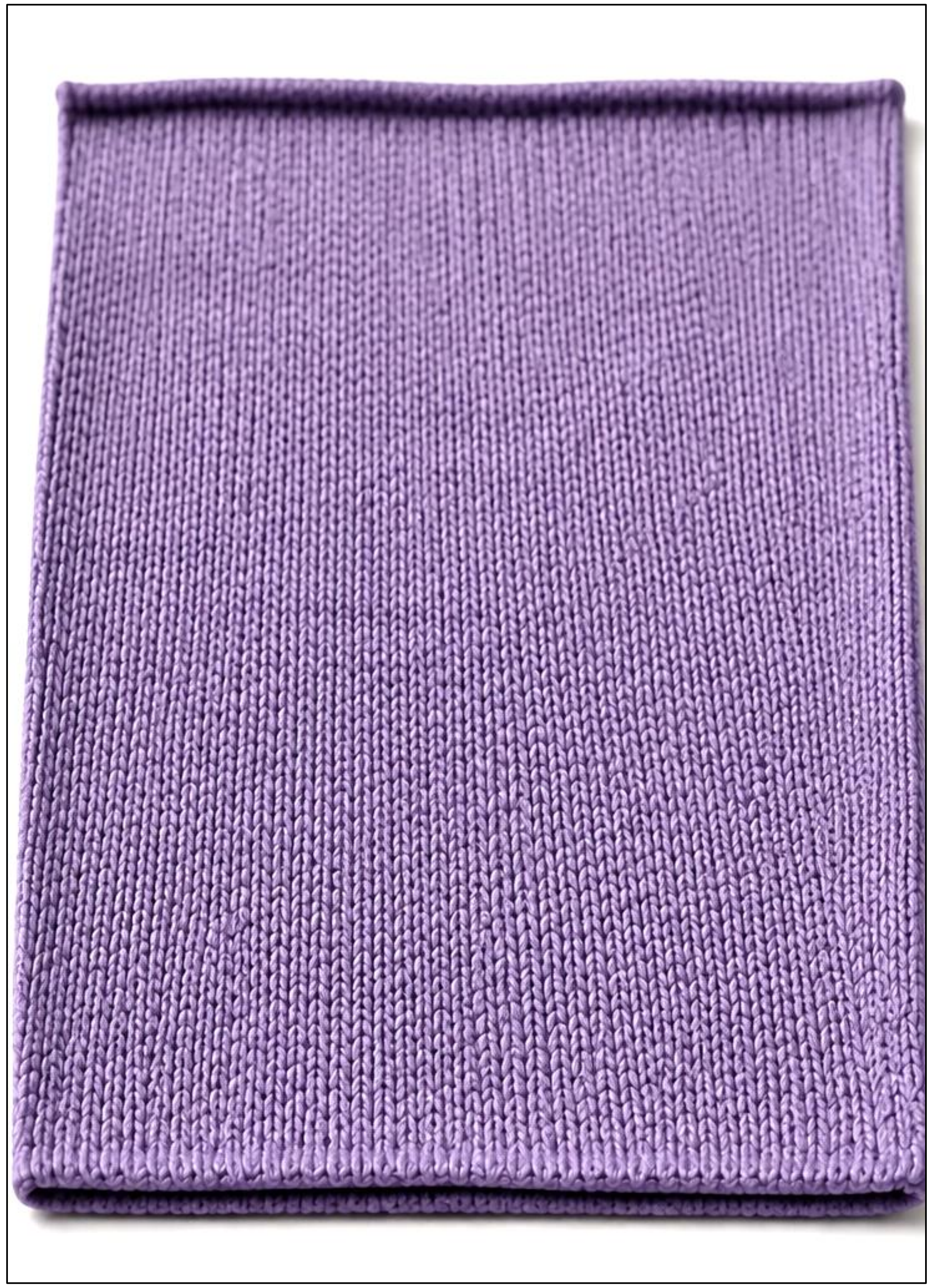
- Fabric length machine
- Garment length machine



FEATURES OF FABRIC LENGTH MACHINE

- Large diameter, circular, latch needle machine knit fabric at high speed.
- The fabric is manually cut away from the machine, usually in roll form, after a convenient length has been knitted.
- Most fabric is knitted on circular machines, either single cylinder (Single jersey) or "Dial and cylinder (Double jersey).
- Unless used in tubular body width, the fabric requires splitting into open width.
- Fabric is finished on continuous finishing equipment and is cut-and-sewn into garments.
- Generally, cam settings and needle set outs are not altered during the knitting of the fabric.
- The productivity, versatility and patterning facilities of fabric machines vary considerably.

FEATURES OF FABRIC LENGTH MACHINE



FEATURES OF GARMENTS LENGTH MACHINE

- It includes straight bar frames, and Circular garment machine.
- These machines are coarser gauge machines than fabric machines.
- They produce both outerwear, and innerwear.
- Garments may be knitted in either tubular or open width form; in the later case, more than one garment panel may be knitted simultaneously across the knitting bed.
- The fabric take down mechanism is adaptable with variable rates of production during the knitting sequence, and on some machines be able to assist both in the setting up on empty needles and take away of separate garments or pieces on completion of the sequence.
- The fabric take down mechanism is more sophisticated than for continuous fabric knitting.

FEATURES OF FABRIC LENGTH MACHINE

NEEDLE GAITING

The way of two beds of knitting needles and their housings are aligned with each other in a dial and cylinder of circular knitting machine or flat bed knitting machine.

Generally, two types of needle gaiting are used. These are:-

- Rib gaiting
- Interlock gaiting

NEEDLE GAITING

1. Rib Gaiting: When the needles in the cylinder and dial in circular knitting machine or front needle bed and back needle bed in flat knitting machine are placed at alternate position with each other is known as Rib gaiting.

Fig: Rib Gaiting

NEEDLE GAITING

2. Interlock Gaiting: Interlock gaiting: When needles in the cylinder and dial in circular knitting machine or front needle bed and back needle bed in flat knitting machine are placed at face-to-face position with each other is known as Interlock gaiting.

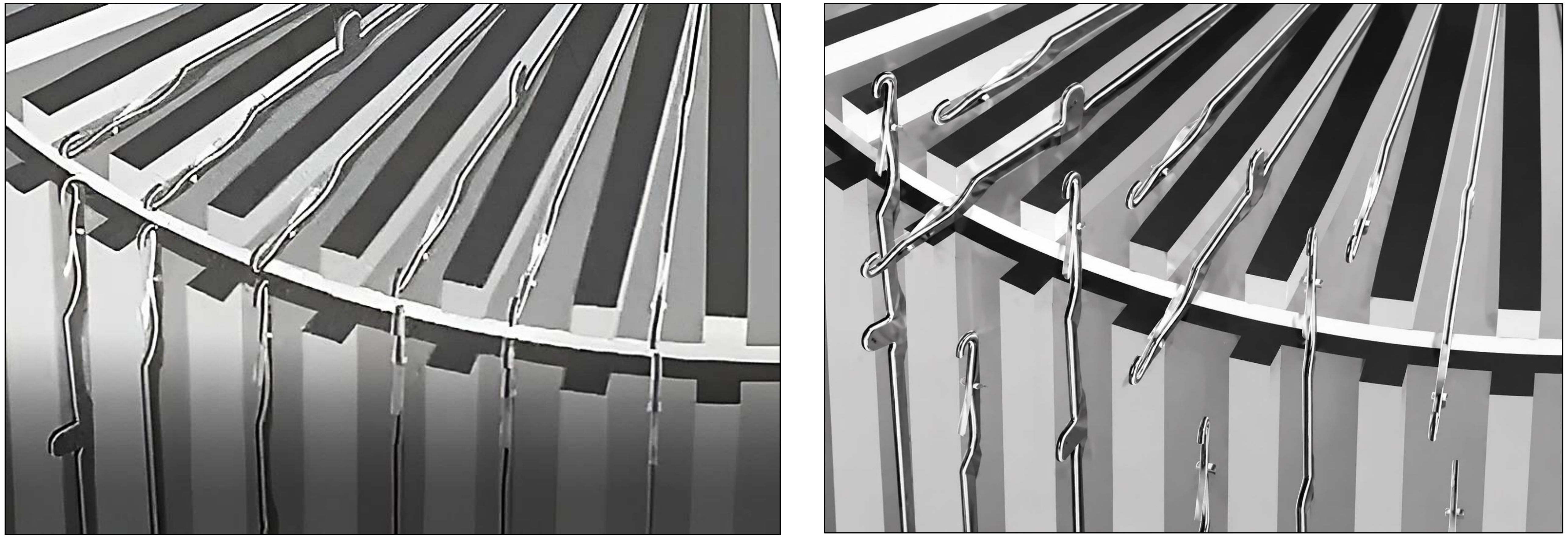
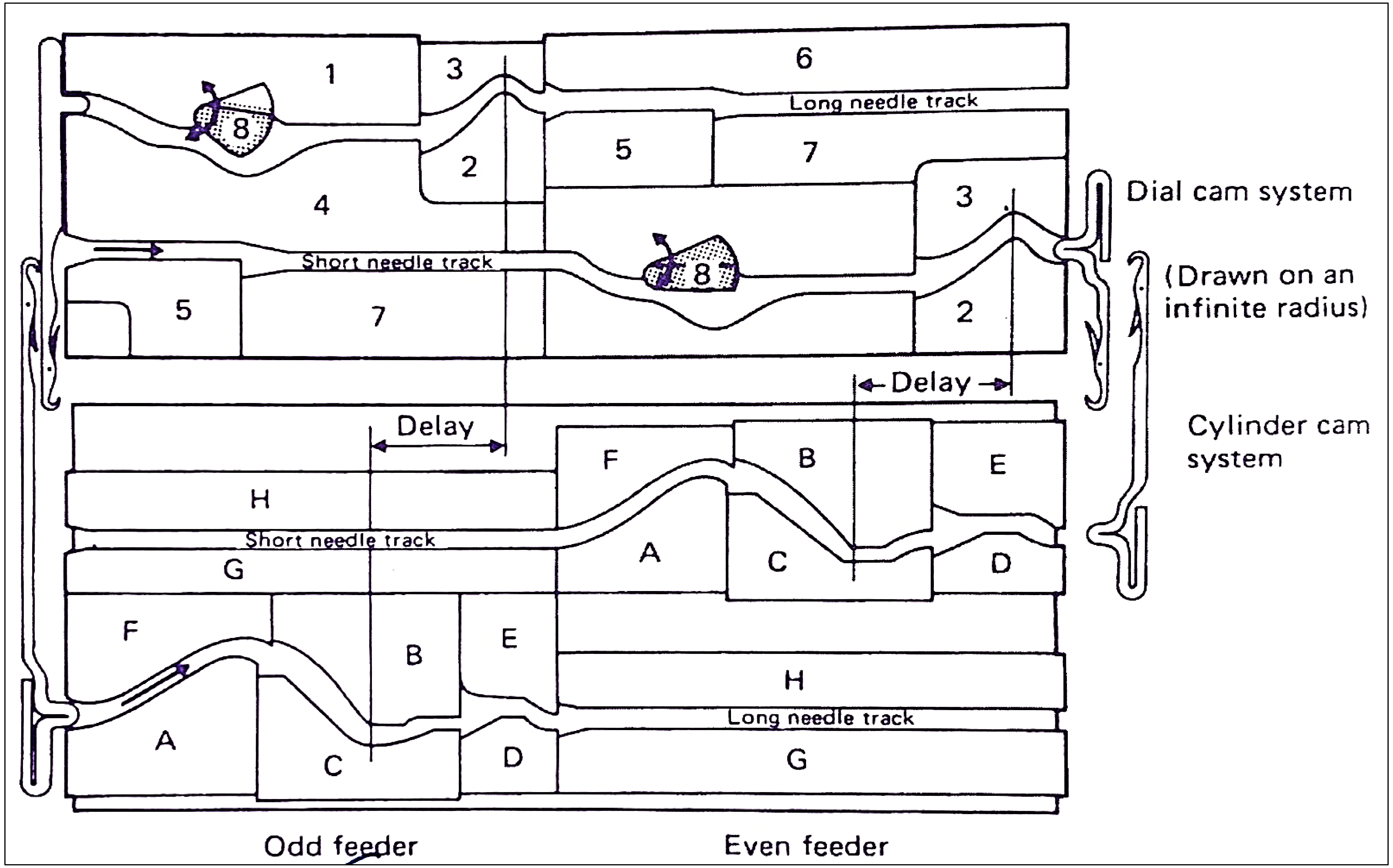


Fig: Interlock Gaiting

NEEDLE GAITING



THANK YOU !